

Performance Work Statement (PWS)

LOCAL AREA NETWORK (LAN)/DEPLOYABLE LOCAL AREA NETWORK (DLAN) ENGINEERING AND INSTALLATION SUPPORT TO NETWORK ENTERPRISE CENTER (NEC)

1 SCOPE

The contractor shall provide personnel, management, and any other items and services to perform LAN/DLAN engineering and installation support to commands within the Pacific Region described below.

1.1 PROJECT PLANNING & ENGINEERING

1.1.1 PROJECT COORDINATION

1.1.2 CONTRACTOR FURNISHED PROPERTY

The contractor shall furnish all equipment and services as necessary to perform the requirements of this contract.

1.1.3 PROJECT DOCUMENTATION

The contractor shall prepare engineering, installation specifics, final installation drawings and as-built drawings in Micro Station SE format or AutoCAD as specified in each task or delivery order. All drawings will be consistent with A/E/C CADD Standard Manual developed by the CADD/GIS Technology Center (CGTC) for Facilities, Infrastructure, and Environment to eliminate redundant Computerized Aided Design and Drafting (CADD) standardization efforts within the Department of Defense (DOD) and the Federal Government. Information may be obtained from the CGTCs web page at: <http://tsc.wes.army.mil>. The number of copies each soft and hard will be as specified in each order will be provided after contract completion and acceptance. The contractor shall prepare and submit testing documentation to determine system, subsystem, or equipment compliance within five (5) days of testing to COR and project engineer or technical staff for review and recommend for acceptance or non-acceptance.

1.2 WORK LOCATIONS

All work will be performed within the Pacific area of responsibility that include but are not limited to: Hawaii, Alaska, and Japan (including Okinawa), Guam and Kwajalein Atoll. Therefore, travel may be required. The requirement to perform these tasks is not all-inclusive but will be specified

in individual task orders (TO). To provide accurate quotes, contractors are encouraged to be present at all site surveys. However, travel expenses are not reimbursable.

1.2.1 LOCAL TECHNICAL SUPPORT

Contractors shall specify if they have local technical support to cover site surveys at all locations in 1.2. In addition, through the life of this contract, contractors shall keep the list of the local support team current.

1.3 TRANSPORTATION

The contractor is responsible for all transportation and travel costs associated with the performance of the order. Transportation costs should be accounted for in the firm-fixed price quote submitted in response to the solicitation.

1.4 CONTRACTOR POINT OF CONTACT

1.4.1 POCs

The contractor shall provide the Contracting Officer (KO), at the post award conference, a local toll-free telephone number for all locations (if applicable), and a designated point of contact(s) to receive calls. The designated point of contact(s) will coordinate contractor technicians in the performance of tasks and response in the allotted time.

1.4.1.1 ELECTRONIC COMMUNICATIONS

The contractor may utilize electronic communication (such as email, cellular phones, pagers, and/or answering machines). Calling a cellular phone/pager or an answering machine will constitute notification by the Government and shall respond no later than 48 hours. The contractor shall notify the KO when changes are made to the designated point of contact(s) or phone numbers.

1.5 TEMPORARY STOP WORK

The KO reserves the rights to temporarily stop work in progress.

1.6 PERIOD OF PERFORMANCE

The period of performance for this contract is specified in the schedule.

1.7 MATERIALS

Contractors are required to provide all documentation required by the Government to validate receipt of materials and services with itemized costs listed in the quotes for the materials and services separated. These will include but not limited to, shipping invoices and bills of materials.

1.8 WARRANTY

Contractor shall warranty all service work (to include repairs) for a minimum of six months. All equipment supplied shall be warranted in accordance with standard manufacturer's warranty. Warranty will start from date of acceptance.

1.9 OPERATION HOURS:

1.9.1 NORMAL HOURS OF OPERATION.

The contractor shall provide engineering support functions during normal duty hours 7:30 a.m. to 4:30 p.m., Monday through Friday, excluding Federal holidays. The contractor shall not perform work after normal duty hour without prior approval of the Contracting Officers Representative (COR).

1.9.2. EMERGENCY AFTER NORMAL DUTY HOURS.

Emergency service response between two (2) and four (4) hours, depending on criticality of repair or safety issue will be approved by the COR. The contractor shall provide emergency contact information POC name and telephone number in the event of an emergency. As tasked by the COR, the contractor shall perform after duty emergency service as follows:

4:30 p.m. through 7:30 a.m. Mondays through Fridays

4:30 p.m. through 7:30 a.m. Fridays through Saturday

7:30 a.m. Saturday through 7:30 a.m. Monday

2. QUALITY CONTROL

The contractor shall have an inspection system covering the requirement of this contract. The adequacy of reports and documentation specified in this requirement shall be the responsibility of the contractor. It will include such reports as stated in the PWS and section 3.6 TECHNICAL DOCUMENTATION of this document and will include weekly progress reports with the current status of projects, problems encountered, resolutions of problems, and point of contacts. All quality control operations performed by the contractor shall be subject to Government verification. Verification can consist of monitoring to determine that the practices and methods of the contractor's procedures are properly applied, review of contractor's quality control findings and results, and conducting audits and reviews of the contractor's operations.

2.1 QUALITY ASSURANCE

All services and deliverables under this contract will be inspected by the COR or a designated representative for acceptance. The Contractor shall employ its commercial quality control procedures in the performance of the requirement of this contract to ensure installation and repair services are performed. Acceptance criteria shall be based upon all deliverable items meeting accepted professional standards for technical content, workmanship, clarity, completeness, and other factors that indicate acceptability as specified by each TO. The government will periodically evaluate the contractor's performance in accordance with the Service Performance Summary (PRS).

2.2 DELIVERABLES

The contractor shall provide deliverables as described in subsequent task orders. Deliverables shall be specified by the government. Format and delivery schedule for deliverables shall be outlined in the solicitation.

Number	Name	Frequency	Quantity
A001	Technical and As-built Documents	Per Task Order (TO)	1
A002	Information Security Documents	Per TO	1
A003	Engineering/installation specifications and instructions	Per TO	1
A004	Test documentation	Per TO	1
A005	Reports, plans, and other narrative deliverables	Per TO	1
A006	Weekly Status Reports	Weekly	1
A007	Quality assurance inspection report	After completion	1
A008	OTDR traces that show power loss	As needed	1
A009	Contract proposal	7 Business days	

3 SPECIFIC TASKS

The contractor shall provide information technology engineering, engineering surveys, wireless surveys, implementation support, and technical documentation. Services and support shall be for outside plant cabling, support structures, internal wiring distribution, and wireless infrastructure installation. Installation Campus Area network (ICAN) and systems and shall be in accordance with DOD and Army regulations and standards.

TOs shall include a separate PWS that defines the terms specific to the order. In the event a PWS conflicts with terms in the basic contract, the basic contract shall guide. The contractor shall engineer, design solutions, and perform installations in compliance with Department of Army Regulations, cited standards and guidance to accomplish the task (see references). The proposal shall submit an optimized solution to the Government's requirement.

Contractor shall develop a quote, work plan, work schedule, and engineering design. With these documents, the contractor shall submit the best solution possible. The plan shall be submitted to the Contracting Officer Representative (COR) by specified date agreed upon following the site survey. The COR, NEC engineer or other technical Government representative shall review the plan for approval.

The contractor shall coordinate use of existing pathways (aerial and underground) with servicing contractor American Telephone and Telegraph Company (e.g. AT&T) and Joint Hawaii Information Transfer System (JHITS) which are telecommunications contracts or follow on service provider or Network Enterprise Center (NEC) cable maintenance shop using established

procedures. The contractor shall provide butterfly drawing for underground conduits and request a pole position on aerial installation. The contractor shall also perform pole line loading analysis and assess pole condition to ensure the integrity and safety of the cable placement. The contractor shall prepare and submit testing documentation to determine system, subsystem, or equipment compliance within five (5) days of testing to the COR, and project engineer or technical staff for review and recommend for acceptance or non-acceptance. Conduct analysis and pole condition to ensure the integrity and safety of the cable placement.

3.1 ENGINEERING

The contractor shall perform engineering to transform the design approach and system requirements into a configuration that can be procured, installed, tested, integrated, and supported within the state of existing technology and supportability. The contractor's engineering process shall incorporate planning and design procedures to develop technical solutions, technical evaluations, and cost estimates and ensures adherence to current United Facilities Criteria (UFC) 3-580-01 (see Appendix).

3.1.1

Engineering shall determine equipment configuration and placement. Include alternating current (A/C) or direct current (D/C) power load requirements, Heating Ventilation and Air conditioning (HVAC) requirements, Global Broadcast System (GBS) requirements, and physical security appropriate for class of network. All drawings shall be to scale.

3.1.2

Engineering shall provide cable, jack assembly assignment plan and drop location IAW: 1) National Standards and UFC 3-580-01, 2) The Contractor shall use the NEC standard labeling practice that will be provided for each individual task order.

3.1.3

Engineering shall determine components required in the installation (bills of material). All network equipment is required to be on the Unified Capabilities Approved Products List (UCAPL) in accordance with the DoDI 8100.04.

3.1.4

Switches and components including labor shall be priced separately.

3.1.5

All engineered solutions shall meet the current version of the Standards and Specifications for the DISN-Enterprise Network Installation Campus Area Network Design and Implementation (DISN-EN ICAN-DI).

3.1.6

Certifications required for the vendors to obtain a DLAN award:

- Fiber Optic Association (FOA) certification
- Certified Wireless Network Administrator (CWNA) or Aruba Certified Mobility

Associate (ACMA) for Wi-Fi installations or Ekahau Certified Solutions Engineer (ECSE) certification for wireless LAN Design

- BISC Registered Communications Distribution Designer (RCDD) certification

Other requirements for vendors to obtain a DLAN award:

- Must be able to obtain a facility security clearance (FCL)
- Must comply with AR389-49
- SIPR requires adherence to CNSSAM TEMPEST / 1-13 and CNSSI No. 7003

3.2 TELECOMMUNICATION ROOM (TR) REMEDIATION

TR Remediation actions for the contractor shall improve network infrastructure by complying with Army and Industry Standards. This includes the following, but is not limited to:

3.2.1

The contractor shall replace, add and / or remove door locks to ensure TR doors and cabinets lock properly. All obsolete door locks and associated hardware / equipment shall be removed. The contractors shall install cabinets and / or racks as required. The contractors shall add, remove, replace cabinets, racks and A/C cooled, wall or floor mounted as required. The contractors shall add, replace, repair various types of HVAC equipment to include wall mount, split, central HVAC systems to ensure properly functioning HVAC equipment.

3.2.2

The contractors shall install, make electrical / lighting adjustments such as replace fixture, light bulbs etc. The contractors shall install dedicated electrical outlets. The contractors shall convert to dedicated electrical outlets or install new dedicated electrical outlets. The contractors shall install L5-20 or L6-30 electrical outlets and define location within the electrical distribution panel(s) for installation / connection of electrical outlets to circuit breakers. The contractors shall install, add, remove and replace UPS and ensure UPS matches and are compatible with the dedicated electrical system. The contractors shall install, add, remove and replace PDU and ensure PDU matches and are compatible with dedicated electrical system. The contractors shall install, add, remove, replace bonding/grounding and ensure bus bar contains 2-hole lug connector. Bonding and grounding installations shall connect to electrical panel, steel, other bus bar grounded rod, etc. and shall comply with NEC standards. The contractors shall install, add, remove 4" PBB, 4" SBB, RBB bus bars and ensure each contains 2-hole, copper IAW with NEC standards. The contractors shall install fire rated backboard. The contractors shall add, replace, remove backboards system to allow for proper bus bars and OSP copper/fiber entry and termination onto the backboards.

3.2.3

The contractors shall label cabinets and racks and ensure proper RU labels IAW TIA/ EIA standards. The contractors shall label the fiber/copper patch panels IAW with 30th Signal Battalion standards. The contractors shall label, dress, and replace patch cords. The contractors shall remove, replace and ensure proper location of communication drops for homeruns.

3.3 WIRELESS

Wireless infrastructure engineering must be complaint with 311th Wireless Local Area Network

CONOPS and DISN-EN NIPR ICAN Wi-Fi Design & Implementation (N-ICAN Wi-Fi-DI) standards. Contractors shall install coverage and WIDS devices to provide WLAN services to the customer.

3.3.1

Wireless installation staff requires a minimum certification of vendor neutral Certified Wireless Network Associate (CWNA) or Aruba Certified Mobility Associate (ACMA). The contractor shall utilize their own Predictive Modeling Tool such as Ekahau Pro to perform coverage maps to install WAP/WIDS devices in the areas efficiently to meet project requirements. The Government requires a Wi-Fi Network Site Survey Report consisting of heat maps from the predictive modeling site survey of the space, Access Point hardware list for the project areas and supporting documents for TEMPEST Countermeasure Review. The Government also requires a post installation RF Survey Heat Map confirming wireless service coverage requirements are met alongside as built drawings and cable test reports.

3.4 (POTENTIAL) HAZARDOUS MATERIAL (HAZMAT) REMEDIATION

HAZMAT Awareness Training required of all personnel involved in the project. This shall cover identification, handling, and proper disposal procedures for hazardous materials that may be encountered on-site. The contractor must minimally follow the requirements listed for remediating HAZMAT:

3.4.1

CERTIFICATIONS AND LICENSES: Verify that contractors hold or can subcontract to vendors holding appropriate certifications and license for handling HAZMAT substances IAW local, state, and federal regulations.

3.4.2

SAFE HANDLING PROCEDURES: Contractors must adhere to safety protocols for handling, transporting, and disposing of HAZMAT materials. This includes providing appropriate personal protective equipment (PPE) for their workers and implementing containments.

3.4.2.1

The Contractor will be responsible for properly managing and disposing of any or all waste generated. Any generation, storage and disposal of hazardous waste must be managed in accordance with USAG HI 200-4 installation Hazardous Waste

3.5 SUPPORT

a. The above telecommunication information provides the mission essential capabilities for combat readiness, security and life safety.

b. In addition, the contractor shall consider standardization and life cycle costs in technical solutions provided as per this document and directed by Army Regulation 70-1.

c. Failure to design a technical solution to meet these essential capabilities and functional requirements can result in mission failure. Failure shall be documented as negative performance in the Contract Discrepancy Report (CDR).

3.5.1

IMPLEMENTATION SUPPORT. The contractor shall install, test, and make operational system components (hardware, software, and firmware). Ensure software with CON, JTIC or UCAPL approved hardware. Provide interoperable systems. 95%.

3.5.2

The contractor shall perform daily work area clean up so that workspaces are left safe and free from obstruction in compliance with safety standard operational procedures.

3.5.3

The contractor shall specify in their proposal and prior to the start of work to the COR if a staging area is required.

3.5.4

All supporting contractor staff will have all required training, certifications, and security clearance to include Baseline Certification and Computer Environment certification commensurate with DoD 8570.01, AR 25-2, NETCOM Regulation 25-5, and DA 380-5. The contractor shall be manufacturer trained and certified (e.g. Holocom) to be able to install a PDS (Protected Distribution System) in accordance with CNSSI No. 7003, ISEC current SIPRNet Technical Implementation criteria. The contractor shall be trained to implement wireless infrastructure compliant with all physical separation requirements specified in applicable Committee on National Security Systems Advisory Memorandum (CNSSAM) Telecommunications Electronics Material Protected from Emanating Spurious Transmissions (TEMPEST) guides.

3.5.5

The contractor is responsible for submitting and obtaining approval of appropriate dig permits, confined space entry permits, roadwork permits, vehicular traffic routing plan, aerial work and climbing permits and other permits as required to perform work.

3.6 TECHNICAL DOCUMENTATION

3.6.1

The contractor shall provide technical documents to the COR for review prior to the final walk through for project acceptance, to include all IA supporting documentation for net worthiness as required by supporting Information Assurance Manager (IAM).

3.6.2

The contractor shall prepare engineering/installation specifications and instructions that depict the configuration of physical layout, rack profiles, electrical interface, interconnections, and other engineering installation requirements such as electrical distribution, HVAC, emergency disconnect, fire suppression, cable distribution hardware (Cable path by ladder or J-hook). The contractor shall provide installation drawings and as-built drawings in Computer Aided Design (CAD) format.

3.6.3

The contractor shall prepare test documentation and shall participate in inspections, tests, or evaluations for, the purpose of, determining system, subsystem, or equipment compliance. Contractor shall identify and perform in accordance with UFC standard. All tests shall meet the UFC standards and manufacturer specifications.

3.6.4

The contractor shall provide three (3) sets of all reports, plans, and other narrative deliverables in hard copy and CD ROM using the current versions of the following software:

Text: Microsoft Word

Spreadsheet: Microsoft Excel

Drawings: AutoCAD (2010) or Micro Station

Sketches: PowerPoint

3.6.5

The contractor will provide weekly status reports to the COR on progress, issues, and work for the coming week.

3.6.6

Prior to project acceptance by the Government, contractor will show the Government inspection team documentation that they have conducted their quality assurance inspection(s) and will provide details of items corrected.

3.7 INSTALLATION SPECIFIC

3.7.1

The contractor shall install, test, troubleshoot, repair, and remove optical fiber cables and copper cables. The contractor shall install and remove aerial and underground support structures. The contractor shall provide all supplies/ materials, labor, equipment and oversee the installation or repair of new communication infrastructure and systems. The contractor shall install dedicated power circuit and receptacle for each rack/cabinet IAW UFC electrical power standards. Power surveys and coordination with DPW (Directorate of Public Works) are required when installing dedicated electrical circuit. Report critical issues with recommended solutions to the COR identified in specific task. Facility access issues shall be coordinated with the COR and servicing S2 to avoid schedule delays.

3.8 OPTICAL FIBER CABLE INSTALLATION (INTERIOR AND EXTERIOR)

3.8.1

The contractor shall install aerial, underground (direct buried, inner duct in conduits or in conduits), and internal to facility optical fiber cable.

3.8.2

The contractor shall splice, terminate, and patch single-mode and/or multi-mode optical fiber cable using passive alignment systems or the active alignment systems IAW UFC criteria.

3.8.3

Contractor is responsible for submitting and obtaining the approval of appropriate dig permits, confined space entry permits, roadwork permits, vehicular traffic routing, aerial work and climbing permits if applicable to the installation work.

3.9 CABLE LABELING SCHEME IAW BASE STANDARDS AND UFC 3-580-01

3.10 OPTICAL FIBER CABLE TROUBLESHOOTING (INTERIOR AND EXTERIOR):

3.10.1

The contractor shall troubleshoot aerial, underground (direct buried, inner duct in conduits or in conduits), and internal to facility optical fiber cable to the termination ends (patch panel, hub, patch/zip cords).

3.10.2

The contractor shall measure the bi-directional loss on optical fiber cable. Test results from actual end-to-end power meter and optical fiber performance tester to include OTDR traces that show power loss, media characteristics and trouble areas or anomalies shall be identified and provided in color printed form to the COR.

3.10.3

The contractor shall notify the COR if they have not identified the problem within four hours after arriving on site. The contractor shall provide status and the estimated time of completion. Notify COR within four hours of arriving at site. Adhere to delivery schedule. 95%.

3.10.4

The contractor shall report to the location designated in the work request within eight working hours after notification.

3.10.5

Contractor is responsible for submitting and obtaining approval of appropriate dig permits, confined space entry permits, roadwork permits, vehicular traffic routing, aerial, and work.

3.11 OPTICAL FIBER CABLE REPAIR (INTERIOR AND EXTERIOR):

3.11.1

The contractor shall repair aerial, underground (direct buried, inner duct in conduits or in conduits), and internal optical fiber cable to the termination ends (patch panel, hub, patch/zip cords) or other items in the circuit path. Ensure UFC Criteria is met.

3.11.2

The contractor shall measure the bi-directional loss on repaired optical fiber cable. Test results from actual end-to-end power meter and optical fiber performance tester to include OTDR traces that show power loss, media characteristics and trouble areas or anomalies shall be identified and provided in color printed form to the COR.

3.11.3

The contractor shall report to the location designated in the work request within eight working hours after notification.

3.11.4

The Contractor shall notify the COR if the repair has not been completed with four hours after arriving on site. The contractor shall provide status and the estimated time of completion, estimated cost for repair, BOM and COA's for temporary and/or long-term restoration.

3.11.5

Contractor is responsible for submitting and obtaining approval of appropriate dig permits, confined space entry permits, roadwork permits, vehicular traffic routing, aerial work and climbing permits if applicable to the repair work.

3.12 COPPER CABLE REMOVAL (INTERIOR AND EXTERIOR):

3.12.1

The contractor shall remove aerial, underground (direct buried, inner duct in conduits or in conduits), and internal to facility copper cable and other cables as identified by the COR.

3.12.2

The contractor shall report to the location designated in the work request. The contractor shall follow the delivery schedule for completion of work. Adhere to delivery schedule. 95%.

3.12.3

The contractor shall palletize the removed cable. The contractor shall deliver the pallet(s) to Building 6042, East Range or other designated location by COR.

3.13 AERIAL AND UNDERGROUND SUPPORT STRUCTURE

3.13.1

The contractor shall install and/or remove aerial and underground support structures to include utility poles, maintenance holes, conduits, pull boxes, hand holes, guying, etc.

3.13.2

The contractor shall follow the delivery schedule for complete work.

3.14 SECURE INTERNET PROTOCOL ROUTER NETWORK (SIPRNET) and NON-SECURE INTERNET PROTOCOL ROUTER NETWORK (NIPRNET)

3.14.1

The contractor shall be required to provide SIPRNET and NIPRNET solutions. The Government reserves the right to provide further clarifications and defining of tasks at time of ordering with Contractor.

3.14.2

The contractor's key personnel, engineers and/or installers shall hold current security clearances (minimum SECRET) to support SIPRNET installations.

3.14.3

Contractor must adhere to ISEC SIPRNet Technical Implementation Criteria to include the approval before installation starts. See Appendix 5.1.

3.15 ELECTRICAL MODIFICATIONS

3.15.1

The contractor shall be required to provide electrical modifications to work locations, which will include new electrical wiring and/or rewiring to provide power to equipment. This work is to be done by a bonded licensed electrician. Electrical modifications require power surveys and shall be submitted for review and approved by DPW.

3.16 BUILDING CABLING SYSTEM LABELING

Labeling of installed cables shall follow current TIA/EIA-606 standards and the local NEC labeling scheme.

3.16.1

The contractor shall provide the KO, at the post award conference, a local telephone number for all locations, and a designated point of contact(s) to receive calls. The designated point of contact(s) will arrange to enable contractor's technicians to receive tasks and respond

4 PERSONNEL.

The Contractor shall provide personnel possessing the skills, knowledge, and training to satisfactorily perform the services required by this contract. Personnel performing work under this work statement shall remain an employee of the Contractor and shall not be considered an employee of the Government.

4.1 MINIMUM QUALIFICATIONS.

4.2 INFORMATION TECHNOLOGY ENGINEER / PROGRAM MANAGER

The engineers shall have a Bachelor of Science (BS) degree in engineering from an accredited

institution and has demonstrated the experience, knowledge, skills and past experience to perform systems integration and engineering tasks in the area of telecommunications or a Professional Engineer (PE) license with a minimum of 5 years of experience. Resume of the Information Technology Engineer shall be included in the proposal. This ability shall be supported with specific examples detailing achievements and results of past performance. Experience shall demonstrate the capability to define requirements for large, corporate-wide, multi-node information systems. A Registered Communications Distribution Designer (RCDD) certification provided by Building Industry Consulting Service International (BICSI) is highly desired. Shall be Certified Wireless Network Associate (CWNA) or Aruba Certified Mobility Associate (ACMA).

4.2.1

PROGRAM MANAGER JOB DUTIES. The Program Manager is a professional who organizes, plans, and executes telecommunication projects while working within restraints like technical items, budgets and schedules. The Program Manager's responsibilities include project planning, team coordination, risk management, budget oversight, client communication and maintaining quality standards. Program Managers lead entire teams, define project goals, communicate with stakeholders, and see a project through to its closure.

4.3 INSTALLATION PERSONNEL.

Installation personnel shall consist of individuals that have graduated from an accredited electronics vocation technical school or equivalent and has demonstrated the experience, knowledge, skills and past experience to perform work specified in the PWS. Resume shall be included in the proposal. Certified BICSI installer highly desired. Shall be Certified Wireless Network Associate (CWNA) or Aruba Certified Mobility Associate (ACMA).

4.3.1

INSTALLER JOB DUTIES. An installation technician completes tasks related to the installation of various types of equipment. Technician shall install and remove equipment and test connections to ensure proper functionality. Installation technicians shall secure tools and bill of materials safely and precisely. Technician shall be responsible for working as a team with all other job personnel to complete each job and task. Technician may be required to read installation schematics, layouts and drawings. Technicians shall maintain safety at all times.

4.4 LICENSED ELECTRICIAN.

The project electricians shall be a licensed and bonded electrician who has demonstrated the experience, knowledge, skills and past experience to perform work specified in the PWS. Resume shall be included in the proposal.

4.4.1

LICENSED ELECTRICIAN JOB DUTIES. An Electrician is a skilled professional who installs, maintains, and repairs electrical systems in buildings and structures. An Electrician executes wiring plans, installs fixtures and equipment, and ensures electrical safety compliance. Electricians hold expertise in various electrical systems, tools, and safety regulations. Duties shall include testing, repairing, maintaining electrical equipment, connecting electrical equipment and appliances as well as applying electrical specifications and electrical theory to determine best solutions to job

requirements. Electrician shall conduct inspections of circuit breakers, transformers, and other electrical components for safety and conformance to national regulatory codes.

4.5

The contractor's key personnel, engineers and/or installers shall hold current security clearances (minimum SECRET) to support SIPRNET installations.

5. APPENDIXES

5.1 APPLICABLE DOCUMENTS FOR SIPRNET INSTALLATIONS

5.1.1 ORDER OF PRECEDENCE

In the event of a conflict between the text of this PWS and the applicable document cited herein, the text of the PWS should take precedence. Nothing in the PWS however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

5.1.1.1 Government / Industry/UFC Standards

5.1.1.1.1 CNSSI No. 7003 September 2015 Protective Distribution Systems (PDS)

5.1.1.1.2 MIL-HDBK-419A Grounding, Bonding, and Shielding Practices and Procedures for Electronic Equipment Facilities

5.1.1.1.3 OSHA 2206 Occupational Safety and Health Administration General Industry Standards

5.1.1.1.4 ANSI C2 National Electrical Safety Code

5.1.1.1.5 NFPA 70 National Electrical Code

5.1.1.1.6 Telecommunications Industry Association TIA/EIA Telecommunications Building Wiring Standards

5.1.1.1.7 ANSI/TIA /EIA-526-14, Measurement of Optical Power Loss of Installed Multimode Fiber Cable Plant

5.1.1.1.8 ANSI/TIA/EIA-569, Commercial Building Standard for Telecommunications Building Wiring Standards

5.1.1.1.9 ANSI/TIA/EIA-568B.1, Commercial Building Telecommunications Cabling Standard

5.1.1.1.10 Manufacturer Requirements for Cable Installation to preserve the installed cable warranty of performance.

5.1.1.1.11 NSI/TIA/EIA-568-B.3, Optical Fiber Cabling Components Standard

5.1.1.1.12 CNSSAM TEMPEST/ 1-13 RED/BLACK Installation Guidance

5.1.1.1.13 EIA/TIA-606 Administration Standard for Commercial Telecommunications Infrastructure

5.1.1.1.14 ANSI/TIA 607 Telecommunications Bonding and Grounding (Earthing) for Customer Premises

5.1.1.1.15 DOD Documents

The issues in effect, of the following documents, on the date of the award, form a part of this PWS to the extent specified herein.

5.1.1.1.16 UFC-3-580-01 Telecommunications Interior Infrastructure Planning and Design

5.1.1.1.17 DISN-Enterprise Network Installation Campus Area Network Design and Implementation (DISN-EN ICAN-DI).

5.1.1.1.18 DISN-Enterprise Network Non-classified Internet Protocol Router Installation Campus Area Network Wi-Fi Design and Implementation (DISN-EN N-ICAN Wi-Fi-DI)

5.1.1.1.19 Army Regulation 25-1 Army Information Technology

5.1.1.1.20 Army Regulation 25-2 Information Assurance

5.1.1.1.21 Army Regulation 380-27 Control of Compromising Emanations

5.1.1.1.22 UFGS-33 82 00 Telecommunications Outside Plant (OSP)

5.1.1.1.23 T-OSP-PDC 21-010 Telecommunications Outside Plant Planning and Design Criteria

5.1.1.1.24 US Army 311th Signal Command Wireless Local Area Network CONOPS

5.1.1.1.25 Unified Capabilities Approved Products List (UCAPL) in accordance with the DoDI 8100.04

5.1.1.1.26 ISEC current SIPRNet Technical Implementation criteria

5.1.1.1.27 USAG HI 200-4 installation Hazardous Waste.

6. ABBREVIATIONS AND ACRONYMS

ACMA	Aruba Certified Mobility Associate
ANSI	American National Standards Institute
APL	Approved Products List
AQL	Acceptable Quality Level
AR	Army Regulations

AT&T	American Telephone and Telegraph Company
BOM	Bill of Materials
CAD	Computer Aided Design
CADD	Computerized Aided Design and Drafting
CGTC	CADD/GIS Technology Center
CAN	Campus Area Network
CD	Compact disc
CDR	Contract Discrepancy Report
CDRL	Contract Data Requirements List
CD ROM	Compact Disc Read-only-memory
COA	Course of Action
CoN	Certificate of Net worthiness
COR	Contracting Officer's Representative
CNSSAM	Committee on National Security Systems Advisory Memorandum
CNSSI	Committee on National Security Systems
CWNA	Certified Wireless Network Associate
DA	Department of the Army
DISN	Defense Information System Network
DISN-EN	Enterprise Network
DLAN	Deployable Local Area Network
DOD	Department of Defense
DoDI	Department of Defense Instruction
DPW	Directorate of Public Works
EIA	Electronic Industries Alliance
GBS	Global Broadcast System, Global Broadcast Service
GIS	Geographical Information System
HVAC	Heating Ventilation and Air Conditioning
IAM	Information Assurance Manager
IAW	In accordance with
ICAN	Installation Campus Area Network
ICAN-DI	Installation Campus Area Network Design and Implementation
ISEC	Information System Engineering Command
JHITS	Joint Hawaii Information Transfer System
JITC	Joint Interoperability Test Command
KO	Contracting Officer
LAN	Local Area Network
MIL HDBK	Military Handbook
NEC	Network Enterprise Center
NETCOM	U.S. Army Network Enterprise Technology Command
NFPA	National Fire Protection Association
NIPRNET	Non-secure Internet Protocol Router Network
NSTISSI	National Security Telecommunications and Information System Security Instruction
NSI	National Security Information.
NSTISSAM	National Security Telecommunications and Information Systems Security Advisory Memorandum TEMPEST
OSHA	Occupational Safety and Health Administration
OSP	Outside plant
OTN	Optical Transmission Network
OTDR	Optical Time Domain Reflectometer

PBB	Primary Bonding Busbar
PDS	Protected Distribution System
PDC	Telecommunications Outside Plant Planning and Design Criteria
POC	Point of Contact
PRS	Performance Requirements Summary
PWS	Performance Work Statement
RBB	Rack Bonding Busbar
SBB	Secondary Bonding Busbar
SIPRNET	Secure Internet Protocol Router Network
STIG	Security Technical Implementation Guides
TEMPEST	Telecommunications Electronics Material Protected from Emanating Spurious Transmissions
TIA	Telecommunication Industry Association
TO	Task Order
TR	Telecommunication Room
UCAPL	Unified Capabilities Approved Products List
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
WAP	Wireless Access Point
WIDS	Wireless Intrusion Detection System

Performance Requirement Summary (PRS)

PERFORMANCE WORK STATEMENT FOR DLAN ENGINEERING AND INSTALLATION SUPPORT

Statements	Standards/AQLs	Incentive/Remedy
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3.1	The contractor shall perform engineering to transform the design approach and system requirements into a configuration that can be procured, installed, tested, integrated, and supported within the state of existing technology and supportability. The contractor's engineering process shall incorporate planning and design procedures to develop technical solutions, technical evaluations, and cost estimates and ensures adherence to current UFC-3-580-01 Telecommunications Interior Infrastructure Planning and Design	a) Meets technical specifications AQL: 95%	Re-perform
3.5.1	The contractor shall install, test, and make operational system components (hardware, software, and firmware). Ensure software with CON, JITIC or APL approved hardware.	a) Provide interoperable systems AQL: 95%	Re-perform
3.8.2	The contractor shall splice, terminate, and patch single-mode and/or multi-mode optical fiber cable using passive alignment systems	a) Compliant with applicable standards, Tech Orders, and manufacturer's manuals AQL: 95%	Re-perform

	or the active alignment systems IAW UFC criteria.		
3.10.3	The contractor shall notify the COR if they have not identified the problem within four hours after arriving on site. The contractor shall provide status and the estimated time of completion.	a) Notify COR within four hours of arriving at site. AQL: 95%	
3.10.4	The contractor shall report to the location designated in the work request within eight working hours after notification	a) Report to site eight hours after notification AQL: 95%	
3.11.3	The contractor shall report to the location designated in the work request within eight working hours after notification.	a) Report to site eight hours after notification AQL: 95%	
3.11.4	The Contractor shall notify the COR if the repair has not been completed with four hours after arriving on site. The contractor shall provide status and the estimated time of completion, estimated cost for	a) Report to site eight hours after notification AQL: 95%	

	repair, BOM and COA's for temporary and/or long-term restoration.		
3.11.5	Contractor is responsible for submitting and obtaining approval of appropriate dig permits, confined space entry permits, roadwork permits, vehicular traffic routing, aerial work and climbing permits if applicable to the repair work.	a) Provide all required permits AQL: 95%	Re-perform
3.12.2	The contractor shall report to the location designated in the work request. The contractor shall follow the delivery schedule for completion of work.	a) Adhere to delivery schedule AQL: 95%	Re-perform